(b) Symmetrical Case:

When $p = q = \frac{1}{2}$, the two point distribution becomes

$$f(x) = \begin{cases} \frac{1}{2} & \text{for } x = x_1 \\ \frac{1}{2} & \text{for } x = x_2 \end{cases}$$

and the cumulative distribution function is

$$F(x) = \begin{cases} 0 & \text{if } x < x_1 \\ \frac{1}{2} & \text{if } x_1 \le x < x_2 \\ 1 & \text{if } x \ge x_2 \end{cases}$$

The mean and variance respectively are $\frac{x_1 + x_2}{2}$ and $\frac{(x_2 - x_1)^2}{4}$.

11.6.3 The Bernoulli distribution



Jacob Bernoulli (1654 - 1705)

Independent trials having constant probability of success p were first studied by the Swiss mathematician Jacques Bernoulli (1654–1705). In his book *The Art of Conjecturing*, published by his nephew Nicholas eight years after his death in 1713, Bernoulli showed that if the number of such trials were large, then the proportion of them that were successes would be close to p.

In **probability theory,** the **Bernoulli distribution**, named after Swiss mathematician **Jacob Bernoulli** is the **discrete probability distribution** of a **random variable**. A Bernoulli experiment is a random experiment, where the outcomes is classified in one of two mutually exclusive and exhaustive ways, say success or failure (example: heads or tails, defective item or good item, life or death or many other possible pairs). A sequence of Bernoulli trails occurs when a Bernoulli experiment is performed several independent times so that the probability of success remains the same from trial to trial. Any nontrivial experiment can be dichotomized to yield Bernoulli model.

Definition 11.10: (Bernoulli's distribution)

Let X be a random variable associated with a Bernoulli trial by defining it as X (success) = 1 and X (failure) = 0, such that

$$f(x) = \begin{cases} p & x = 1 \\ q = 1 - p & x = 0 \end{cases} \qquad \text{where } 0 < p < 1 \text{ ,}$$

then X is called a Bernoulli random variable and $f(x)$ is called the Bernoulli distribution.

Or equivalently

If a random variable X is following a Bernoulli's distribution, with probability p of success can be denoted as $X \sim Ber(p)$, where p is called the parameter, then the probability mass function of X is

$$f(x) = p^x (1-p)^{1-x}, \qquad x = 0, 1$$

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